

Skewness

$$a = [2, 3, 3, 4, 5, 6, 7, 8, 25]$$

Now

$$\text{Mean } (\bar{x}) = \frac{2+3+3+4+5+6+7+8+25}{9}$$

$$= \frac{63}{9} = 7$$

$$\text{Median} = \frac{N+1}{2} = \frac{9+1}{2} = \frac{10}{2} = 5^{\text{th}} \text{ term}$$

$$\therefore \text{Median} = 5$$

Now, Standard Deviation

x	$x - \bar{x}$	$(x - \bar{x})^2$
2	-5	25
3	-4	16
3	-4	16
4	-3	9
5	-2	4
6	-1	1
7	0	0
8	1	1
25	18	324
$\Sigma(x - \bar{x})^2$		396

$$\text{Variance} = \frac{\Sigma(x - \bar{x})^2}{n}$$

$$= \frac{396}{9}$$

$$= 44$$

$$SD = \sqrt{\text{variance}}$$

$$= \sqrt{44}$$

$$= 6.64$$

$$\text{Skewness} = \frac{3(\text{mean} - \text{mode})}{SD}$$

$$= \frac{3(7 - 5)}{6.64}$$

$$= 0.905 \text{ (Positive or right skewness)}$$

b. $b = [1, 15, 16, 17, 18, 19, 20, 21, 22]$

Here,

$$\begin{aligned}\text{Mean}(\bar{X}) &= \frac{1+15+16+17+18+19+20+21+22}{n} \\ &= \frac{149}{9} = 16.56\end{aligned}$$

$$\text{Median} = \frac{N+1}{2} = \frac{9+1}{2} = 5^{\text{th}} \text{ term}$$

$$\therefore \text{Median} = 18$$

$$\bar{x} = 16.56$$

Now

x	$x - \bar{x}$	$(x - \bar{x})^2$
1	-15.56	242.12
15	-1.56	2.44
16	-0.56	0.32
17	0.44	0.19
18	1.44	2.08
19	2.44	5.96
20	3.44	11.84
21	4.44	19.72
22	5.44	29.60
$\Sigma(x - \bar{x})^2$		314.27

$$\begin{aligned} \text{Now, variance} &= \frac{\Sigma(x - \bar{x})^2}{n} \\ &= \frac{314.27}{9} \\ &= 34.92 \\ &= \underline{\underline{5.91}} \end{aligned}$$

$$\begin{aligned} \text{SD} &= \sqrt{\text{Variance}} \\ &= \sqrt{5.91} = \sqrt{34.92} \\ &= 2.44 = 5.91 \end{aligned}$$

$$\text{Skewness} = \frac{3(\text{mean} - \text{median})}{\text{SD}}$$

$$= \frac{3(16.56 - 18)}{2.44 \cdot 5.91}$$

$$= 0.731 \quad (\text{negative or left skewness})$$

$$C = [10, 12, 14, 15, 15, 15, 16, 18, 20]$$

Now,

$$\text{Mean}(\bar{X}) = \frac{10+12+14+15+15+15+16+18+20}{9}$$

$$= \frac{135}{9} = 15$$

$$\text{Mode} = 15$$

$$\text{Median} = \left(\frac{N+1}{2} \right)^{\text{th}} = \frac{9+1}{2} = \frac{10}{2} = 5^{\text{th}} \text{ term}$$

$$\therefore \text{Median} = 15$$

$$\text{Mean} = \text{median} = \text{mode}$$

Now, SD

x	$x - \bar{x}$	$(x - \bar{x})^2$
10	-5	25
12	-3	9
14	-1	1
15	0	0
15	0	0
15	0	0
16	1	1
18	3	9
20	5	25
$\Sigma(x - \bar{x})^2$		70

$$\text{Variance} = \frac{\Sigma(x - \bar{x})^2}{n}$$

$$= \frac{70}{9}$$

$$= 7.78$$

$$\Rightarrow \text{SD} = \sqrt{\text{variance}}$$

$$= \sqrt{7.78}$$

$$= 2.78$$

$$\text{Skewness} = \frac{\text{mean} - \text{mode}}{\text{SD}}$$

$$= \frac{15 - 15}{2.78}$$

$$= 0 \text{ (symmetric)}$$

Kurtosis

$q = [4.2, 5.1, 4.8, 5.5, 4.9, 5.0, 5.3, 4.7, 5.2, 4.6, 5.4, 4.5, 5.9, 5.0, 5.2, 4.8, 5.3, 4.7, 5.1]$

Now,

$$\begin{aligned} \text{Mean}(\bar{x}) &= \frac{4.2 + 5.1 + 4.8 + 5.5 + 4.9 + 5.0 + 5.3 + 4.7 + 5.2 + 4.6 + 5.4 + 4.5 + 4.9 + 5.0 + 5.2 + 4.8 + 5.3 + 4.7 + 5.1}{20} \\ &= \frac{99.4}{20} = 4.97 \end{aligned}$$

Deviations

x	$x - \bar{x}$	$(x - \bar{x})^4$
4.2	-0.77	0.3515
5.1	0.13	0.000286
4.8	-0.17	0.000835
5.5	0.53	0.0789
4.9	-0.07	0.00024
5.0	0.03	0.000001
5.3	0.33	0.01186
4.7	-0.27	0.00531
5.2	0.23	0.00280
4.6	-0.37	0.01879
5.4	0.43	0.03419
4.5	-0.47	0.04880
5.1	0.13	0.000286
4.9	-0.07	0.000024
5.0	0.03	0.000001
5.2	0.23	0.00280
4.8	-0.17	0.000835

5.3	0.33	0.01186
4.1	-0.27	0.00531
5.1	0.13	0.000286
$\Sigma(x-\bar{x})^4$		0.5745

Divide by n

$$\text{variance} = \frac{\Sigma(x-\bar{x})^2}{n}$$

$$= \frac{0.5745}{20} = 0.0287$$

$$\begin{aligned} \text{SD} &= \sqrt{\text{variance}} \\ &= \sqrt{0.0287} \\ &= 0.3132 \end{aligned}$$

$$\begin{aligned} \text{SD}^4 &= (0.3132)^4 \\ &= 0.00962 \end{aligned}$$

$$\begin{aligned} \text{Kurtosis} &= \frac{1}{n} \frac{(\Sigma x - \bar{x})^4}{\left(\frac{1}{n} (\Sigma x - \bar{x})^2 \right)^2} \\ &= \frac{0.0287}{0.00962} \end{aligned}$$

$$= 2.98 \text{ (Mesokurtic normal distribution)}$$

6. 5.0, 5.1, 4.9, 5.0, 5.05, 4.95, 5.0, 5.02, 4.98, 5.0, 5.01, 4.99, 5.0, 2.0, 8.0, 1.5, 8.5, 5.0, 5.0, 5.0

$n = 20$

$$\bar{x} = \frac{5.0 + 5.1 + 4.9 + 5.0 + 5.05 + 4.95 + 5.0 + 5.02 + 4.98 + 5.0 + 5.01 + 4.99 + 5.0 + 2.0 + 8.0 + 1.5 + 8.5 + 5.0 + 5.0 + 5.0}{20}$$

$= \frac{99}{20} = 4.95$

Variance =
Average Fourth Power
 $\frac{462.7632}{20} = 23.1382$

Deviations

x	$x - \bar{x}$	$(x - \bar{x})^2$
5.0	0.05	0.00000625
5.10	0.15	0.00050625
4.90	-0.05	0.00000625
5.0	0.05	0.00000625
5.05	0.10	0.00010000
4.95	0.00	0.00
5.0	0.05	0.00000625
5.02	0.07	0.00002401
4.98	0.03	0.00000081
5.0	0.05	0.00000625
5.01	0.06	0.0001296
4.99	0.04	0.00006256
5.0	0.05	0.00000625
2.0	-2.95	75.73350
8.0	3.05	86.53650
1.5	-3.45	141.69950
8.5	3.55	158.82300
5.0	0.05	0.00000625
5.0	0.05	"
5.0	0.05	"

SD
variance = $\frac{92.5770}{20}$
 $= 2.12885$
SD = $\sqrt{2.12885}$
 $= 1.4953$
SD⁴ = $(1.4953)^4$
 $= 4.5400$

kurtosis = $\frac{23.1382}{4.5400}$
 $= 5.10$

$C = [3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 3.2, 3.8, 4.2, 4.8, 5.2, 5.8, 6.2, 6.8, 7.2]$

$n = 18$

$\bar{x} = \frac{3.0 + 3.5 + 4.0 + 4.5 + 5.0 + 5.5 + 6.0 + 6.5 + 7.0 + 3.2 + 3.8 + 4.2 + 4.8 + 5.2 + 5.8 + 6.2 + 6.8 + 7.2}{18}$

$$= \frac{90.2}{18} = 5.011$$

Sum of Squares

$$\sum (x - \bar{x})^2 = 93.1903$$

$$\text{Average} = \frac{93.1903}{2}$$

$$= 5.1772$$

Deviations

$$\text{Variance} = \frac{30.5368}{18}$$

$$= 1.6965$$

$$SD = \sqrt{1.6965}$$

$$= 1.3025$$

$$SD^4 = (1.3025)^4$$

$$= 2.8782$$

$$\text{Kurtosis} = \frac{5.1772}{2.8782}$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$

$$= 1.80$$